

Manifest-Level Benchmark Catalog (N = 100,000)

Global Corpus: 50,000 Benign (5 Subfamilies) + 50,000 Malware (5 Families)

Preserved via Cryptographic SHA-256 Manifest & Metadata Index

Materialization Pipeline
(10 Shards × 1,000 Samples)

Materialized Multimodal Benchmark Cohort (N = 10,000)

Complete Tri-Modal Tensors: Opcode (L=2048) + CFG Graphs + Syscall Sequences

Balanced Distribution: 5,000 Benign (50.0%) | 5,000 Malware (50.0%)

Training Partition (70%)

N = 7,000 Samples

Stratified Training Split

3,500 Benign | 3,500 Malware

(5 Benign + 5 Malware Families)

Validation Partition (15%)

N = 1,500 Samples

Model Selection Split

750 Benign | 750 Malware

(Hyperparameter Tuning)

Held-Out Test Partition (15%)

N = 1,500 Samples

Final Evaluation Split

756 Benign | 744 Malware

(Zero-Leakage Benchmark)

Evaluation Split Protocol & Provenance Assurance

- Zero Data Leakage: Disjoint SHA-256 partition with strictly verified zero hash overlap across splits.
- WannaCry Holdout: 157 sample-level held-out ransomware instances verified in the 1,500 test partition.
- Materialization Integrity: 10,000 tri-modal tensors pre-computed into deterministic PyTorch shards.